

Data Science Assignment

Customer Churn Prediction

Business Problem

You are a Data Scientist at a telecommunications company. The company wants to identify customers who are likely to **churn** so that the retention team can proactively engage with them.

Using the provided **Telco Customer Churn dataset**, build an end-to-end machine learning solution to predict customer churn.

Dataset: IBM Telco Customer Churn Dataset

Target Variable: Churn (Yes / No)

1. Data Understanding & Preparation

Load and investigate the dataset.

Perform appropriate:

- Data type and structure checks
- Missing-value analysis
- Duplicate analysis
- Numerical and categorical feature identification
- Target-variable analysis
- Data cleaning and preprocessing
- Encoding of categorical variables
- Train/Test Split: Use a 70:30 split for training and testing.
- Random Seed: Set random_state = 42 for reproducibility.

Document important observations and explain key preprocessing decisions.

Important: Avoid data leakage and ensure that preprocessing can also be applied consistently to new/unseen data.

2. Exploratory Data Analysis

Perform EDA to understand customer behaviour and factors associated with churn.

Create **at least 5 meaningful visualizations**, covering appropriate numerical and categorical variables.

Your analysis should include:

- Churn distribution
- Customer/service characteristics
- Churn against important customer attributes
- Numerical variable distributions/relationships

For each major visualization, provide a brief business insight.

3. Feature Engineering

Create **at least 2 meaningful features** that could potentially improve churn prediction.

For each feature, explain:

- How it was created
 - Why it may be useful
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4. Model Development

Build a **Decision Tree Classifier** to predict Churn.

You must:

- Train the model on the training dataset.
 - Experiment with **at least two different Decision Tree configurations**.
 - Compare the models.
 - Select and justify the final model.
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5. Model Evaluation

Evaluate the final model using:

- Accuracy
- Precision
- Recall
- F1 Score
- Confusion Matrix

Interpret the results from a **business perspective**.

Also explain:

For a telecom company trying to identify customers who may churn, would you prioritize Precision or Recall? Why?

6. Model Interpretation

Explain what drives the model's churn predictions.

Include:

- Feature importance
- Top features influencing churn
- Decision Tree visualization or appropriate interpretation

Provide a short explanation of the key findings.

7. Model Saving & API

Save the final model/preprocessing pipeline so that it can be reused for new customer data.

Build a REST API using **Flask** or **FastAPI**.

Required Endpoint

POST /predict

The endpoint should:

1. Accept customer information as JSON.
2. Apply the required preprocessing.
3. Load/use the trained model.
4. Return the churn prediction.
5. Return churn probability.
6. Handle invalid input appropriately.

Example response:

```
{  
  "prediction": "Yes",  
  "churn_probability": 0.82  
}
```

8. Submission Requirements

Submit:

- **Jupyter Notebook** — complete analysis and modelling
- **Python/API code**
- **Saved model/pipeline**
- **requirements.txt**
- **README.md** with setup and execution instructions
- **Sample API request and response**

Suggested structure:

```
customer_churn_project/
├── data/
├── notebook/
│   └── churn_analysis.ipynb
├── model/
│   └── churn_model.pkl
├── app.py
├── requirements.txt
├── README.md
└── sample_request.json
```

Expected Outcome

Your final submission should demonstrate the complete workflow:

Business Problem → Data → Preparation → EDA → Feature Engineering → Model → Evaluation → Interpretation → Saved Model → API

Note: Bonus marks may be awarded for additional activities such as **hyperparameter tuning, handling class imbalance, comparing additional models, or other meaningful improvements to the solution.**